

Populations: Limits to Growth
(Read Ch. 2.2; pgs. 46-51)

Name: Key

1. What is a limiting factor? is any factor (biotic or abiotic) that controls the growth of a pop.

2. Limiting factors determine the carrying capacity of an environment for a species.

3. **Support/Reject:** Limiting factors are only biotic factors because they put limits on living things.

Support or Reject this claim AND give your evidence as to why

Reject. Water availability can limit a plants growth and/or survival & water is abiotic.

4. Go to your PearsonRealize dashboard/homepage and complete the assigned interactive; Limiting Factors

Skip slide #4 where it mentions tiles, as the images have trouble loading.

5. Using the table below:

KNOW EACH

- List the density dependent & density independent limiting factors. Describe each factor.
- Give a scenario of HOW it can be a limiting factor

Limiting Factor (density dependent)	Descriptions	Scenario
Competition	↑ in individuals in area means less of a resource or gets used up faster.	Mother feeding baby birds but only has food for one. That goes to the more aggressive chick.
Parasitism	Weaken their host & cause more from host to host. stress	fleas on a dog. When dogs get close fleas will jump from dog to dog.
Disease	Bacteria/Virus can spread in population dense areas.	Disease/unhealthy spaces more common in dense cities rather than rural areas.
Stress from overcrowding	overcrowding can cause species to fight due to stress. Neglect offspring.	Too many fish in an aquarium will cause small/weaker fish to not feed = Death.
Predation	Predators can control a populations size, especially ones that don't have other predators	Great whites eat sea lions controlling numbers.
Herbivory	Herbivores eating too much a specific plant can limit food availability.	Moose eating too much Balsam fir
Limiting Factor (density independent)	Description	Scenario
Environmental extremes	affect all populations regardless of population size	Wildfires will burn all plants & causes all animals to leave a habitat regardless of how many.